

Modelling a Modelling Community: Mapping and Simulating Agent-Based Research in Behavioural Social Science

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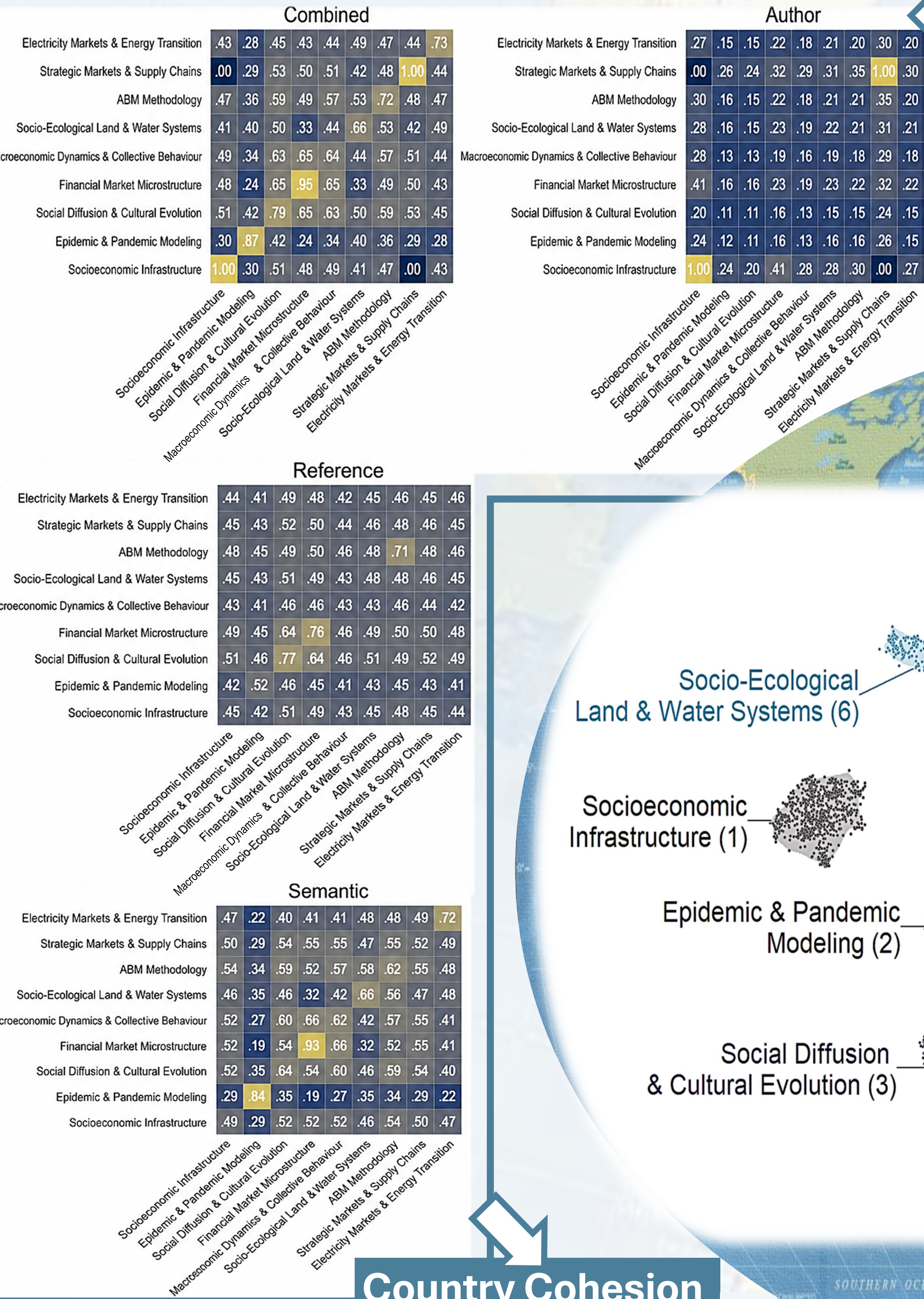
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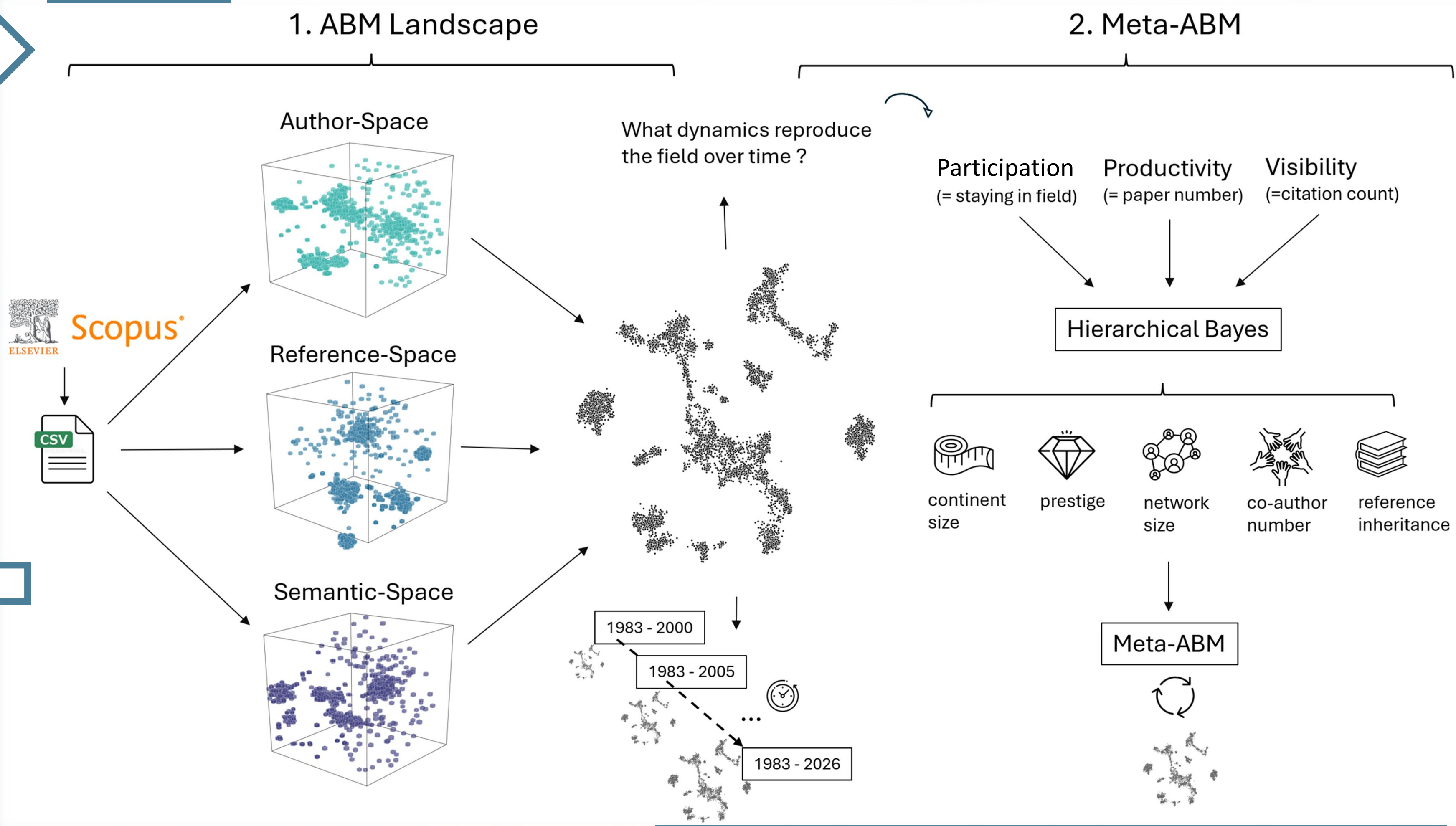
Background

- We lack a systematic map of how **Agent-based Modelling (ABM)** research on human behaviour is organized across domains and how its subfields connect or diverge
- Existing reviews commonly focus on specific applications or narrow methodological strands^{2,3,4}
- To address this gap, we use an LLM-based bibliometric framework¹ (Thoma et al., 2025), to combine semantic, authorship, and reference patterns and map the intellectual landscape of ABM research across the behavioural and social sciences from 1983 to 2026 → **ABM landscape**
- Temporal Bayesian models test **what drives participation** (remaining in the field), **productivity** (number of papers), and **visibility** (number of citations) by testing ‘continent’ prestige, collaboration, team size, and reference reuse as predictors

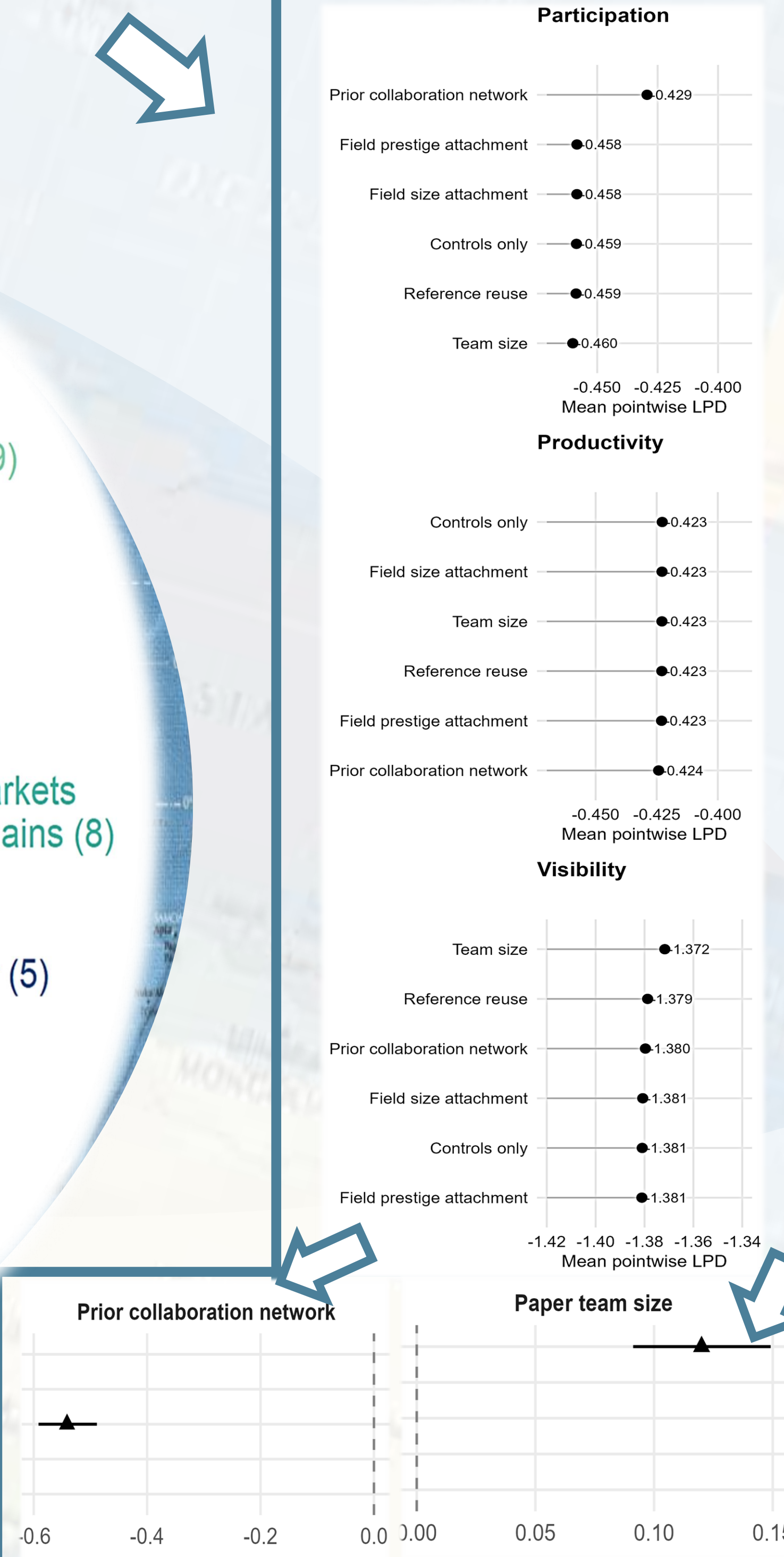
Article Pairs With Above-median Similarity



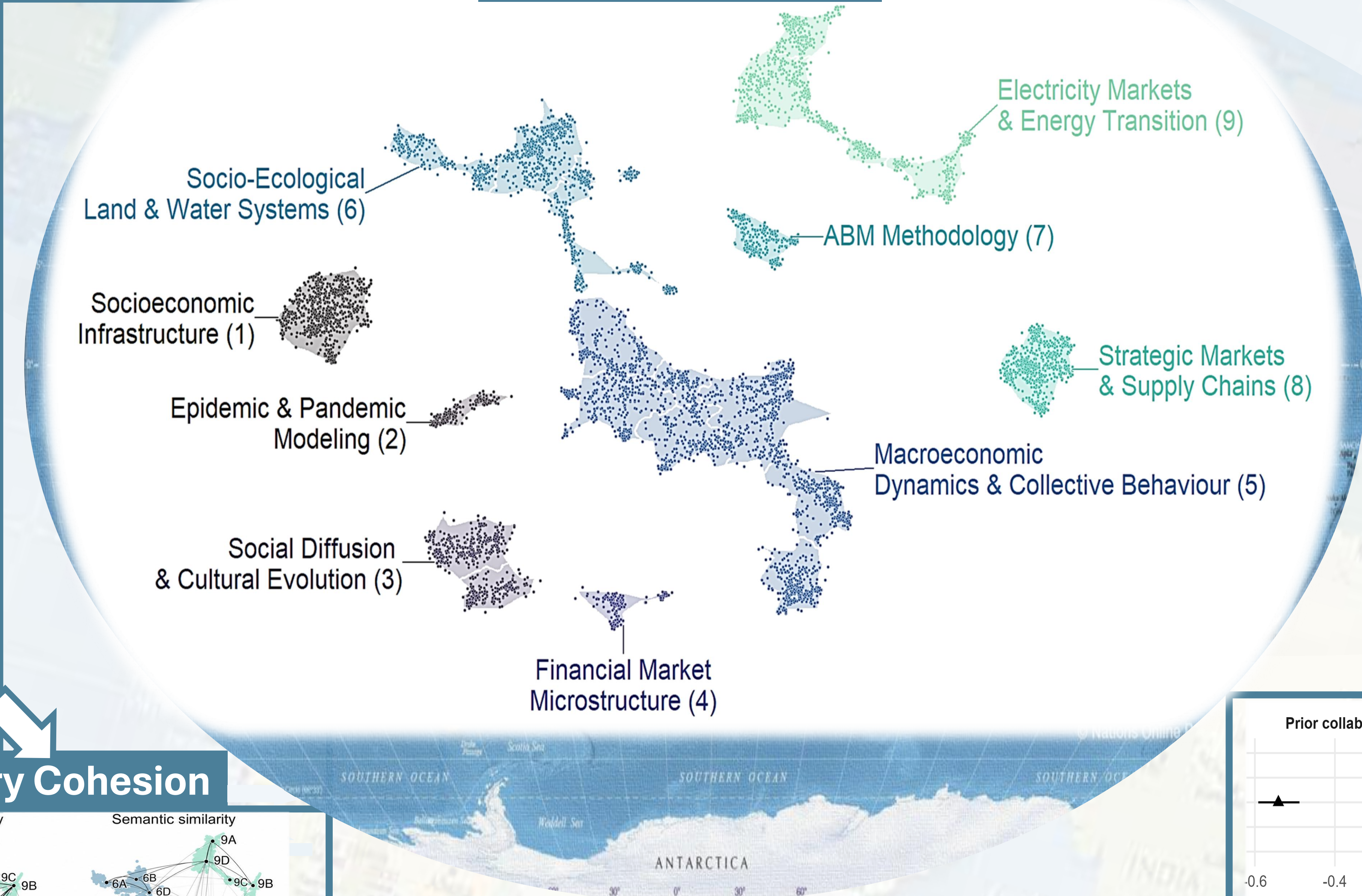
Design



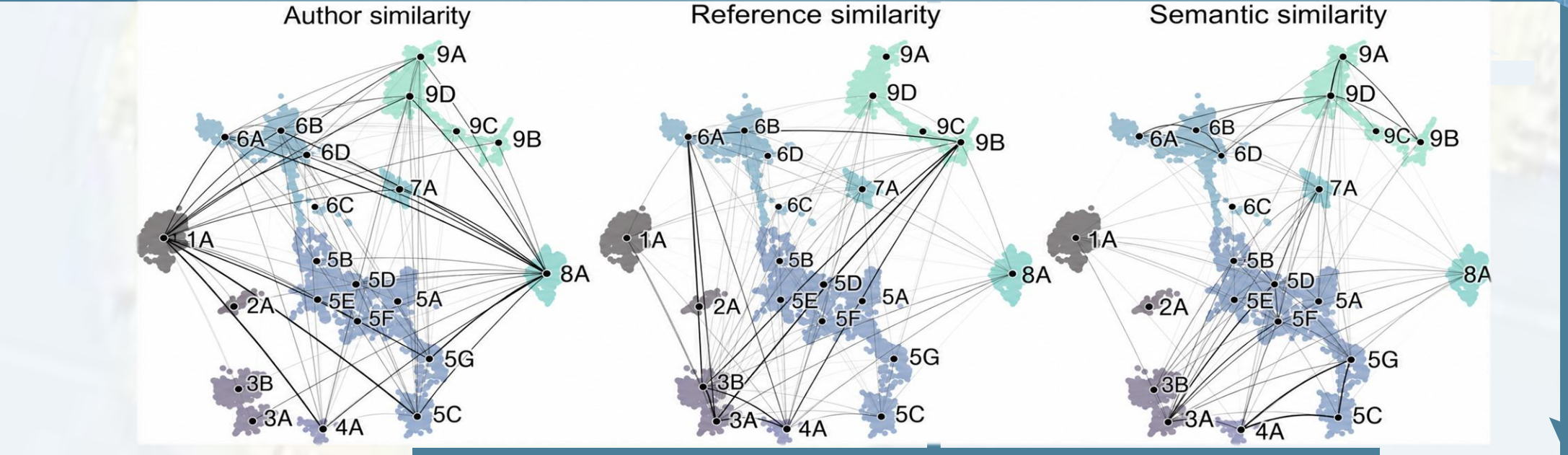
What Drives Scientific Performance ?



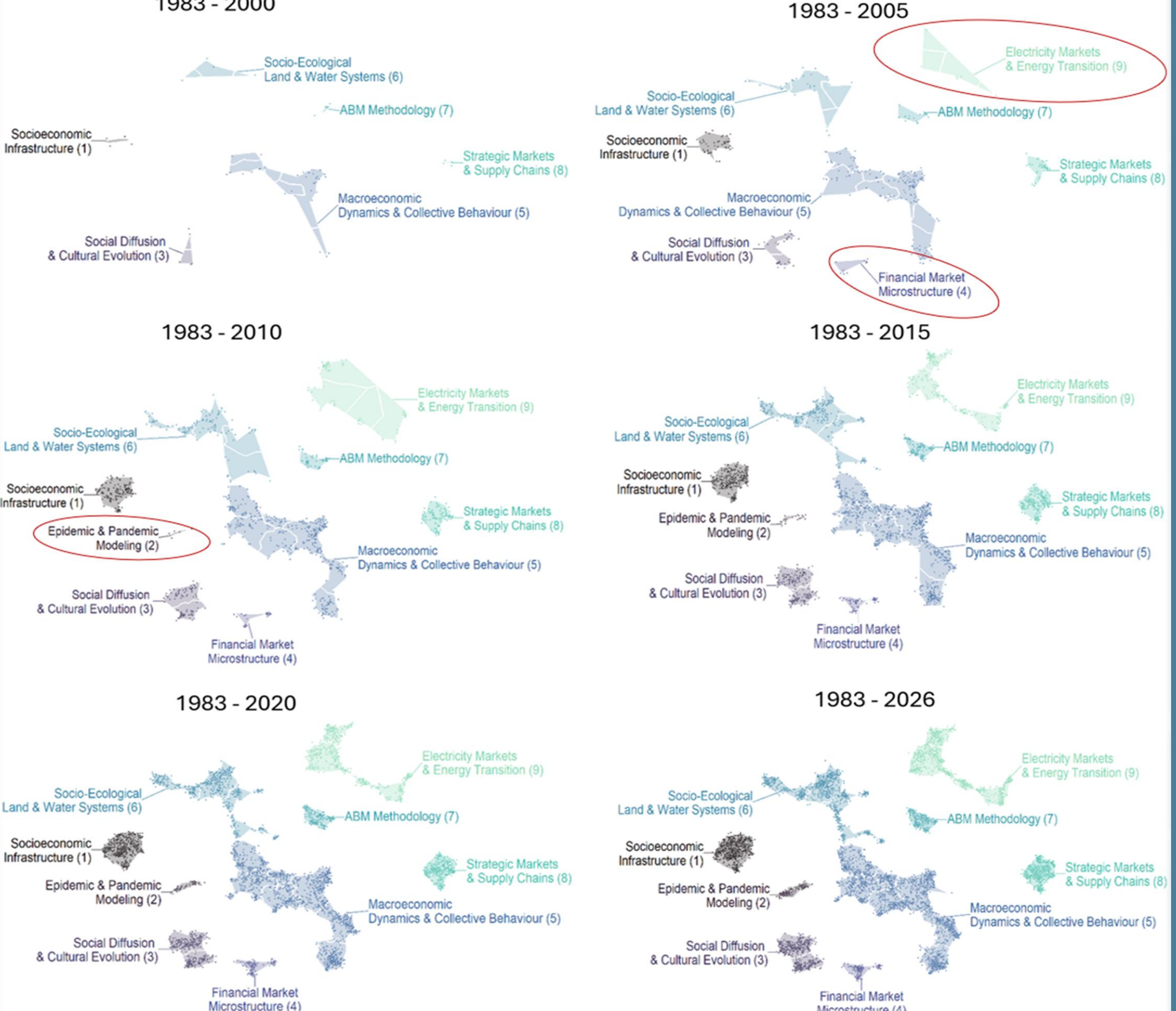
ABM Landscape



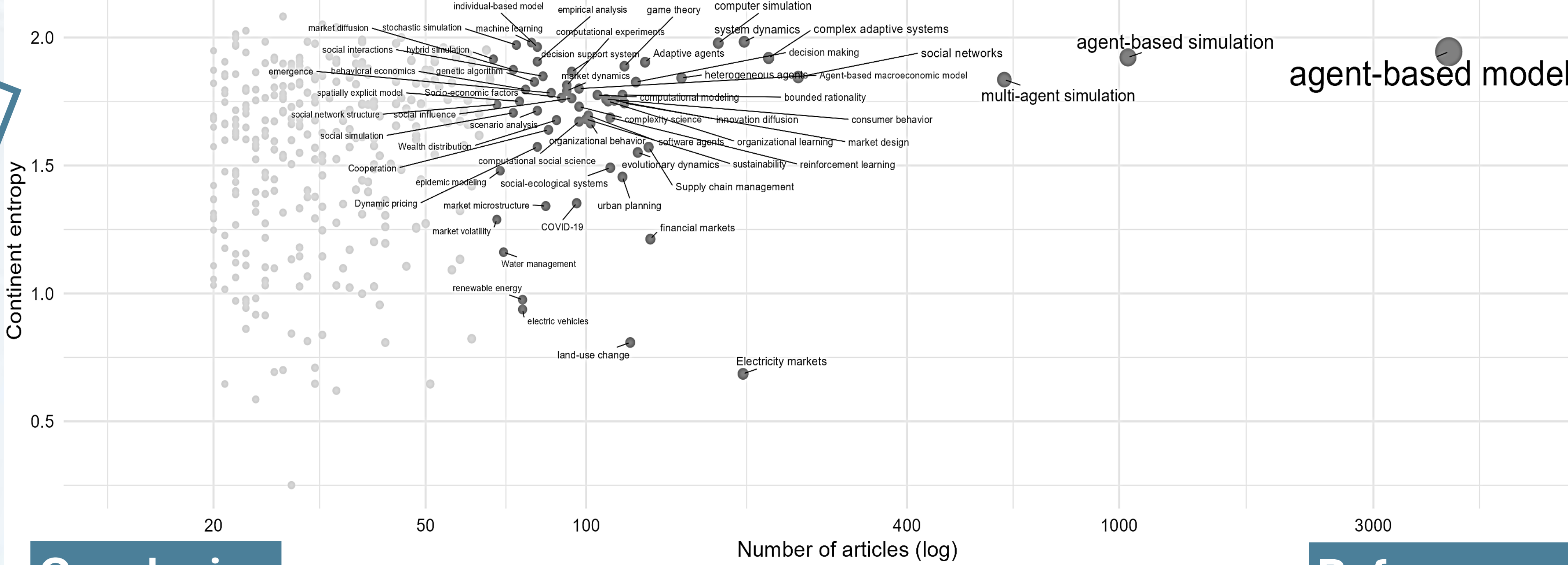
Country Cohesion



ABM Growth (1983-2026)



Tag Prevalence & Geographic Concentration



Conclusion

- Behavioural ABM forms clear thematic regions, but these regions are only weakly integrated across authors, references, and institutions
- New ABM areas emerge as societal issues gain importance, such as energy transition and pandemic modelling
- Papers with more co-authors tend to be more visible (receive more citations) in behavioural ABM → even in a field that studies social systems, the production of science itself is socially organized
- LLM-assisted classification, embeddings, and expert interpretation can help synthesize large research areas more efficiently than traditional review methods alone

References

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